MILITARY SPECIFICATION

COUPLERS, ANTENNA CU-942( )/AR, CU-1894( )/AR AND ANTENNA AS-1703( )/AR

1. SCOPE

1.1 This specification covers the following components, which operate throughout the range of 30.00 to 75.95 megacycles, (See 6.1 and 6.3):

Antenna, Communication AS-1703( )/AR
Coupler, Antenna CU-942( )/AR
Coupler, Antenna CU-1894( )/AR

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids (or request for proposal) form a part of the specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-B-636	Box, Fiberboard
PPP-F-320	Fiberboard, Corrugated and Solid, Sheet Stock (Container Grade) and Cut Shapes
PPP-P-291	Paperboard, Wrapping, Cushioning
PPP-T-45	Tape, Gummed, Paper, Reinforced and Plain, for Sealing and Securing
PPP-T-97	Tape, Pressure Sensitive Adhesive, Filament Reinforced

MILITARY

MIL-P-116	Preservation, Methods of
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-M-13231	Marking of Electronic Equipment
MIL-F-14072	Finishes for Ground Signal Equipment

FSC 5821

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STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-108	Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-147	Palletized and Cantinerized Unit Loads 40 in. x 48 in. Pallets, Skids, Runners, or Pallet Type Base
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-810	Environmental Test Methods

DRAWINGS

SIGNAL CORPS

SC-DL-381491	Antenna, Communication AS-1703( )/AR
SC-DL-381499	Coupler, Antenna CU-942( )/AR
SC-DL-158494	Coupler, Antenna CU-1894( )/AR

(Copies of specifications, standards, drawings and publications required by supplier in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Description. - The Communication Antenna AS-1703( )/AR and Antenna Couplers CU-942( )/AR and CU-1894( )/AR are intended for use with radio sets when installed in fixed or rotary wing aircraft as applicable. The antenna is a fixed length fiberglass whip matched by the antenna coupler to operate throughout the frequency range of 30.00 to 75.95 MHz. The antenna coupler(s) is a broadband, passive circuit unit without any moving elements.

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3.2 Construction. - The equipment shall be constructed in accordance with the applicable drawings as specified on the following drawing and data lists:

SC-DL-381491	Antenna Communication AS-1703( )/AR
SC-DL-381499	Coupler, Antenna CU-942( )/AR
SC-DL-158494	Coupler, Antenna CU-1894( )/AR

3.2.1 Performance. - The coupler(s) on order shall be subjected to and meet the requirements of this specification.

3.2.2 Size and weight. - The size of the coupler(s) shall be as specified on the equipment drawings and the total weight of the coupler(s) (less the fiberglass housing) shall not exceed 14 oz.

3.3 Preproduction samples. - The contractor shall furnish four (4) preproduction samples for approval, if required in the bids and contract. (See 4.3 and 6.2)

3.4 Finish, protective. - The equipment shall be given protective finish in accordance with Specification MIL-F-14072 and the equipment drawings. This includes finish of hardware, screws, etc., and necessary tough-up after mounting. The final paint film shall be Color Number X-24087 in accordance ith MIL-F-14072. (See 4.4)

3.5 Marking. -

3.5.1 General. - Marking shall conform to Specification MIL-M-13231. (See 4.4)

3.5.2 Visibility. - Wherever practicable, parts shall be so mounted that their identification markings will be readily visible with minimum disassembly of the equipment.

3.5.3 Serial numbers. - The following equipments shall be serial numbered: CU-942( )/AR and CU-1894( )/AR.

3.6 Cleaning. -

3.6.1 Parts. - After fabrication, parts shall be cleaned in accordance with good commercial practice or as specified on the equipment drawings. Cleaning processes shall have no deleterious effect. Corrosive material shall be removed completely before the parts are mounted on chassis, panels, etc.

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3.6.2 Units. - After assembly equipment shall be cleaned thoroughly and shall be free from particles of solder, flux and other foreign material. In addition, when necessary, such cleaning shall also be performed before final assembly of the equipment.

3.7 Alignment (parts). - All movable parts shall operate properly without binding or other undue restriction. Parts shall not be assembled in a misaligned condition.

3.8 Securing of parts. -

3.8.1 General. - Brackets, lugs, inserts, bolts and other mounting arrangements shall retain items securely when subjected to specified service conditions.

3.8.2 Securing of parts by threaded fasteners. - Nylok or other screws with plastic devices and Loctite or similar sealants shall not be subjected to temperatures in excess of 250°F (as during baking of paint). Loctite or similar sealants shall be applied in accordance with the manufacturer's instructions.

3.9 Soldering. -

3.9.1 Solder. - Solder shall conform to Requirement 5 of MIL-STD-454.

3.9.2 Acid or acid salts. - No acid or acid salts shall be used in preparation for or during soldering; however, exception is permitted for preliminary tinning of electrical connections and for tinning or soldering of mechanical joints not used to complete electrical circuits, but in no case shall acid or acid salts be used where they can come in contact with insulation material. Where acid or acid salts are used, as permitted above, they shall be completely neutralized and removed immediately after use.

3.9.3 Process. - There shall be no sharp points or rough surfaces resulting from insufficient heating. The solder shall feather out to a thin edge, indicating proper flowing and wetting actions, and shall not be crystalized, overheated, or underheated. The minimum necessary amount of flux and solder shall be used for electrical connections. Any means employed to remove an unavoidable excess of flux shall not incur the risk of loose particles of flux, brush bristles, or other foreign material remaining in the equipment; flux being spread over a larger area; or damage to the equipment. Insulation material that has been subjected to heating during the soldering operation shall be undamaged and parts fastened thereto shall not have become loosened.

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3.10 Tropicalization of materials. - Materials shall be treated as follows:

3.10.1 Treating materials. - Treating materials containing a mercury-bearing fungicide shall not be used. The contractor shall determine that the treating material is compatible with the material or surface to be treated. Selection of material will be held to the practical minimum.

3.10.2 Toxicity. - Treating materials shall cause no skin irritation or other injury to personnel handling the treated material during fabrication, transporation, operation, or maintenance of the equipment, or during use of the finished items used for the purpose intended.

3.10.3 Flexibility. - Treatment shall not affect the flexibility of treated materials, to the extent that the equipment may fail to meet specified requirements when subjected to specified service conditions.

3.10.4 Statement of treatment. - The contractor shall submit for approval to the contracting officer, a statement describing in detail the materials to be treated and the treating materials and processes that he proposes to use. (See 6.2)

3.11 Service conditions (See 4.8). - The equipment shall meet the service conditions below as specified in Specification MIL-E-5400 for Class 1 equipment, except as otherwise specified herein.

3.11.1 Temperature (See 4.8.1). -

a. Operating. - Operable conforming to limits specified from -65 to +132F. Operable from +132F to +150F and no measurement shall deteriorate by more than a factor of 2. The equipment shall be considered Class 1 except when otherwise specified. Fifty (50) watts of RF input power shall be maintained during the operating tests.

b. Non-operating. - Exposure of 24 hours at -65F, and 4 hours at +150F.

3.11.2 Altitude (See 4.8.2). -

3.11.3 Relative humidity (See 4.8.3). - Up to 100 percent relative humidity including conditions wherein condensation takes place in the form of water. This test shall be performed over a 10 day cycle.

3.11.4 Spray-tight protection (See 4.8.4). - The equipment shall meet the Table I test conditions of MIL-STD-108 for "spray-tight" enclosure. There shall be no evidence of moisture inside the equipment.

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3.11.5 Vibration (See 4.8.5). - The equipment shall meet the vibration tests of Specification MIL-T-5422, Curve IV.

3.11.6 Shock (See 4.8.6). -

3.11.6.1 Equipment (See 4.8.6.1). - The equipment shall meet full specification performance after the shock test.

3.11.6.2 Bench handling (See 4.8.6.2). - The equipment shall be subjected to and meet the bench handling test specified.

3.11.7 Orientation, operating and non-operatig (See 4.8.7). - Indefinite storage in any orientation.

3.11.8 Fungus (See 4.8.8). - The equipment shall show no evidence of viable fungus or corrosion after being subjected to 28 days of fungus test in accordance with MIL-STD-810.

3.11.9 Salt fog (See 4.8.9). - The equipment shall be exposed, in normal operating configuration, to salt fog for 48 hours in accordance with MIL-STD-810 and shall show no evidence of corrosion as evidenced by any visible degradation that can be attributed to flaky, pitted, blistered or otherwise loosened finish.

3.12 Antenna element AS-1703( )/AR requirements (See 4.9). -

3.12.1 Radiation element (See 4.9.1). - The length of the radiating element (conductor) shall be within the length specified on the equipment drawing.

3.12.2 VSWR, antenna (See 4.9.2). - The antenna with coupler, Antenna CU-942( )/AR or CU-1894( )/AR shall have a VSWR of 3:1 or less over the frequency range of 30 to 75.95 MHz, when mounted on a four (4) foot diameter ground plane and connected to either Receiver Transmitter RT-348( )/ARC-54, Receiver Transmitter RT-823( )/ARC-131 or AN/ARC-114( ). The impedance of the antenna and coupler(s) shall be nominally 50 ohms over the frequency range of 30 to 75.95 MHz.

3.12.3 Flexure (See 4.9.3). - The antenna element shall not indicate any mechanical or electrical performance degradation after and during subjection to six (6) bends.

3.13 Couplers, Antenna CU-942( )/AR and CU-1894( )/AR requirements. - (See 4.10)

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3.13.1 VSWR coupler (See 4.10.1). - The coupler(s) mounted on the specified ground plane with AS-1703 or equal installed shall have a VSWR of 3:1 or less over the frequency range of 30 to 75.95 MHz. The impedance shall be nominally 50 ohm over the frequency range.

3.13.2 Flexure (See 4.10.2). - With Antenna AS-1703( )/AR, or equal mounted on the coupler(s), the coupler(s) shall not indicate any mechanical or electrical degradation or damage after and during the six (6) bends of the antenna.

3.13.3 Power capability (rf) (See 4.10.3). - The coupler shall be capable of carring an RF power level of not less than 50 watts continuously.

3.14 Vibration preconditioning (See 4.6). - The couplers on order shall be capable of meeting the requirements herein without subsequent processing, after subjection to the vibration preconditioning.

3.15 Operating inspection (See 4.11). - Equipments on order shall be subjected to the operational inspection as specified, after applicable processes, inspections are completed, to determine that each equipment is operable prior to packaging.

3.16 Interchangeability. - Like units, assemblies, subassemblies, and replaceable parts shall be physically and functionally interchangeable without modification of such items or of the equipment (See 4.13). Individual items shall not be hand-picked for fit or performance; however matched pairs or sets, when permitted, may be interchangeable as such. Reliance shall not be placed on any unspecified dimension, rating, characteristic, etc.

3.17 Technical literature and spare parts. - Technical literature and spare parts shall be furnished as specified in the contract. They shall be identical to corresponding parts in the equipment furnished on the order. (See 6.2)

3.18 Workmanship. - The equipment shall be manufactured and assembled in accordance with the equipment drawings (see 3.2) and applicable portions or the following requirements and paragraphs herein:

Soldering	Requirement 5 of MIL-STD-454
Workmanship	Requirement 9 of MIL-STD-454
Cleaning	3.6
Alignment (parts)	3.7
Securing of parts	3.8
Soldering	3.9
Tropicalization	3.10

4. QUALITY ASSURANCE PROVISIONS

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4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection. - Inspection shall be classified as follows:

a. Preproduction inspection (does not include preparation for delivery). (See 4.3)

b. Inspection covered by subsidiary documents. (See 4.4)

c. Quality conformance inspection:

(1) Quality conformance inspection of equipment before preparation for delivery. (See 4.5)

(2) Quality conformance inspection of preparation for delivery. (See 4.14 and 4.15)

4.3 Preproduction inspection. - This inspection will be performed by the Government unless otherwise specified in the contract. It shall consist of the inspection specified in the subsidiary documents covering the items listed in 4.4, and the inspection specified for Group A, Group B and Group C (see Tables I and II and III respectively). The preproduction inspection will normally be performed in this order: (1) vibration, (2) shock, bench-handling and (3) shock, equipment, other preproduction inspection may precede, follow, or be interspersed between the foregoing.

4.4 Inspection covered by subsidiary documents. - The following shall be inspected under the applicable subsidiary documents as part of the inspection before preparation for delivery:

Item	Where Required
Finish	3.4
Marking	3.5

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4.5 Quality conformance inspection of equipment before preparation for delivery. - The contractor shall perform the inspection specified in 4.4 and 4.5.1 through 4.5.6. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition, the Government -- at its discretion -- may perform all or part of the specified inspection, to verify the contractor's compliance with specified requirements (see 6.8). Test equipment for Government verification inspection shall be made available by the contractor.

4.5.1 Group A inspection. - Each unit of each lot of equipment shall be inspected for conformance to all the inspection and test requirements of Table I. Each lot will be subject to verification utilizing the procedures of MIL-STD-105 using the general inspection levels and the AQL's indicated in Table I. Group A inspection shall be performed in any order which is satisfactory to the Government, except that the operational inspection (4.11) shall be last. (See 6.4)

Table I - Group A Inspection

Inspection	Req Para	Insp Para	AQL Major	AQL Minor
Visual & Mechanical:	3.18	4.12		
Coupler CU-942( )/AR (less cover)			0.65%	2.5%
Coupler CU-1894( )/AR (less cover)			0.65%	2.5%
Coupler CU-942( )/AR (complete assembly)			1.0%	4.0%
Coupler CU-1894( )/AR (complete assembly)			1.0%	4.0%
Antenna AS-1703( )/AR			1.0%	4.0%
Electrical:				
Antenna:				
Flexure	3.12.3	4.9.3		
Couplers CU-942( )/AR & CU-1894( )/AR:			*1.0% for the combined group	
VSWR Coupler	3.13.1	4.10.1		
Power Capability	3.13.3	4.10.1		
Operational Inspection	3.15	4.11	0.1%	

*All electrical defects are considered major.

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4.5.2 Group B inspection. - This inspection, including sampling, shall conform to Table II and to the procedures for small sample inspection of MIL-STD-105 using the special inspection levels. The AQL shall be 6.5 percent defective and the inspection level shall be S4 for normal, tightened and reduced inspection. Group B inspection shall normally be performed on inspection lots that have passed Group A inspection and on samples selected from units that have been subjected to and met the Group A inspection.

4.5.3 Order of inspection within Group B. - Group B inspection shall be performed in any order which is satisfactory to the Government.

Table II - Group B Inspection

Inspection	Req. Para	Insp Para	AQL
Antenna AS-1703( )/AR			
Radiation Element	3.12.1	4.9.1	6.5%
Couplers CU-942( )/AR and CU-1894( )/AR			
Flexure	3.13.2	4.10.2	6.5%
Interchangeability	3.16	4.13	6.5%

4.4.5 Group C inspection. - This inspection shall be as listed in Table III and shall be performed on sample unit(s) that have been subjected to and met Group A and Group B inspections. No defects shall be allowed. Sample unit(s) shall be selected in accordance with 4.5.4.1.

Table III - Group C Inspection

Inspection	Rqt Para	Insp Para
Subgroup I:		
Antenna AS-1703( )/AR:		
Temperature	3.11.1	4.8.1
Humidity	3.11.3	4.8.3
Vibration	3.11.5	4.8.5
Couplers CU-942( )/AR and CU-1894( )/AR:		
Temperature	3.11.1	4.8.1
Altitude	3.11.2	4.8.2
Humidity	3.11.3	4.8.3
Vibration	3.11.5	4.8.5

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Table III - Group C Inspection (Cont'd)

Inspection	Rqt Para	Insp Para
Subgroup II:		
Couplers CU-942( )/AR and CU-1894( )/AR:		
Spray-tight protection	3.11.4	4.8.4
Shock	3.11.6	4.8.6
Subgroup III:		
Couplers CU-942( )/AR and CU-1894( )/AR:		
Fungus	3.11.8	4.8.8
Salt Fog	3.11.9	4.8.9

4.5.4.1 Sampling for inspection. - Units shall be selected at random for each group of inspections in Table III as specified in 4.5.4.1.1, through 4.5.4.1.3. The first samples selected shall be at the start of the contract from the first quality conformance inspection production lot. Selection of these samples shall be as specified and within 30 calendar days after selection these samples shall have been subjected to and passed the Group C inspections. (See 6.3)

4.5.4.1.1 Subgroup I. - For this subgroup, two units from the first 25 production units shall be selected for the first Group C inspection. For subsequent Group C inspections, two per month, shall be selected.

4.5.4.1.2 Subgroup II. - For this subgroup, two units (these units may be the units selected for above subgroup I) from the first 25 production units shall be selected for the first Group C inspection. For subsequent Group C inspection, two units every three months, shall be selected.

4.5.4.1.3 Subgroup III. - For this subgroup, two units from the first 25 production units shall be selected for the first Group C inspection. For subsequent Group C inspection, two units shall be selected when one-half of the units on order have been accepted by the Government QA representative.

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4.5.5 Non-compliance. - The contractor shall immediately report in writing each Group C failure occurence, including details of the failure and characteristics affected. The contractor shall immediately investigate the cause of failure and further report the results of investigation and details of the proposed corrective action on (i) the process and materials, as applicable, and (ii) all units of product which were manufactured under the same conditions and which the Government considers subject to the same failure. Reports shall be forwarded to the responsible technical activity designated in the contract through the Quality Assurance Representative. After corrective action has been taken, additional sample units shall be subjected to Group C inspection (all inspections, or the inspections which the sample failed, at the option of the Government) and Groups A and B inspection may be reinstituted; however, final acceptance and shipment will be withheld until the Group C reinspection results have shown that the corrective action was effective.

4.5.6 Reinspection of conforming Group C sample units. - Unless otherwise specified, sample units which have been subjected to and passed Group C inspection may be delivered on the contract or order, provided all damage is repaired, cleaned of all foreign matter, and the sample units are resubjected to and pass Group A inspection.

4.6 Vibration preconditioning. - Place the coupler on a vibration table and operate the system with vibration in a vertical direction between 25 and 55 Hz at an amplitude of 5G's. The Coupler shall be subjected to this preconditioning for one minute.

4.7 General test conditions. - Unless otherwise specified herein, all inspections shall be performed at room temperature, pressure and humidity under the following standard conditions:

a. Input level: RF 50 watts.

b. Ground plane: 4 foot diameter

4.8 Service condition tests. - The service condition tests shall be conducted in accordance with MIL-T-5422 for Class I equipment as defined in Specification MIL-E-5400. Electrical tests shall be run to verify compliance with the applicable paragraphs 3.12.3, 3.13.1 and 3.13.2 at carrier frequencies of 30.50, 50.40 and 75.95 MHz.

4.8.1 Temperature. -

a. Operating. - From -65°F to +132°F

b. Non-operating. - From -65°F to +150°F

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4.8.2 Altitude. -

a. Operating. - Up to 50,000 feet at temperatures as specified in MIL-T-5422 and 3.14.2.

b. Non-operating. - Up to 50,000 feet

4.8.3 Humidity. - The equipment shall be subjected to the humidity test of Specification MIL-T-5422.

4.8.4 Spray-tight protection. - The equipment shall be mounted in its normal operating position and subjected to the tests specified in Table I of referenced specification for compliance with paragraph 3.11.4.

4.8.5 Vibration. - The coupler(s), with AS-1703 Antenna, or equal shall be subjected to the vibration test of MIL-T-5422 for compliance with Curve IV. Tests will be performed before, during and after vibration to verify compliance with paragraphs 3.12.3, 3.13.1 and 3.13.2 at carrier frequencies of 30.50, 50.40 and 75.95 MHz.

4.8.6 Shock. -

4.8.6.1 Equipment. - The equipment shall be subjected to the shock tests of Specification MIL-T-5422.

4.8.6.2 Bench handling. - The coupler shall be removed from its enclosure, as for servicing, and placed in a suitable position for servicing on a solid two-inch fir bench top. The tests shall be performed as follows, in a manner simulating shocks liable to occur during servicing: Tilt up the assembly through an angle of 30 degrees, using one edge of the assembly as an axis, and permit the assembly to drop back freely to the horizontal. Repeat, using other practicable edges of the same horizontal face as axis for a total of four drops. Tests shall be performed before and after shock to verify compliance with paragraphs 3.13.1 and 3.13.2 at carrier frequencies of 30.50, 50.40 and 75.95 MHz.

4.8.7 Orientation, operating and non-operating. - The equipment shall meet the requirements of 3.11.7 when subjected to the conditions of paragraphs 3.12.3, 3.13.1 and 3.13.2.

4.8.8 Fungus. - The equipment shall be subjected to and meet 28 days of fungus testing as specified in Method 508 and MIL-STD-810. For gasket or other nonhermetically sealed equipment, the parts and materials of the chassis shall be sprayed. The equipment shall be tested in this open condition as specified for compliance with 3.11.8.

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4.8.9 Salt fog. - The equipment shall be exposed in its normal operating condition to salt fog in accordance with Method 509 of MIL-STD-810 for 48 hours. The unit(s) shall not be opened and shall not be required to operate. The equipment shall meet the requirement of 3.11.9.

4.9 Antenna Element AS-1703( )/AR tests. -

4.9.1 Radiation element. - The internal conductor of the element shall be measured to determine that its length is as specified on the equipment drawing for compliance with paragraph 3.12.1.

4.9.2 VSWR, antenna. - The antenna element with appropriate coupler shall be mounted on the ground plane to simulate the applicable aircraft, and the VSWR shall be measured at each half megahertz of the frequencies specified for compliance with 3.12.2.

4.9.3 Flexure. - The antenna element shall be placed on its coupler or suitable fixture that will simulate normal operating position. Sufficient weight shall be applied to the tip of the antenna element causing it to deflect into an arc with the tip on a horizontal level with the rigid support of the antenna element. The distance between the tip and the support, when the element is in full arc position, shall be 24 inches. A continuity check shall be conducted before flexure, when the element is in full arc position, and after the flexure test, to ascertain that continuity exists between the tip and base of the element. The element shall meet the requirement of 3.12.3.

4.10 Couplers, Antenna CU-942( )/AR and CU-1894( )/ARC-54, tests. -

4.10.1 VSWR, coupler. - Connect the coupler to the specified ground plane and measure the VSWR at each 0.50 MHz position from 30 to 75.95 MHz for compliance with paragraph 3.13.1

4.10.2 Flexure. - The coupler, with antenna (or equal) mounted, shall be placed in its normal operating position. Sufficient weight shall be applied to the tip of the antenna element causing it to deflect into an arc with the tip on a horizontal level with the top of the coupler cover. The distance between the tip and the cover, when the element is in full arc position, shall be 24 inches. No electrical or mechanical damage shall occur to the coupler(s). The requirements of 3.13.2 shall be met.

4.10.3 Power capability (rf). - Terminate the output of the coupler(s) and apply a carrier frequency at a level of 50 watts for 30 minutes. The coupler shall meet the requirement of paragraph 3.13.3.

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4.11 Operational inspection. - Feed a modulated signal of sufficient amplitude to the input of the coupler. Visually observe the signal by use of an oscilloscope. There shall be no interruption or noise in the visually displayed signal on any one of five (5) frequencies when the coupler is vibrated at 5g's, 25 to 55Hz.

4.12 Visual and mechanical inspection. The equipment shall be examined for the defects listed in Standard MIL-STD-252.

4.13 Inspection for interchangeability. - The dimensions listed below shall be gaged or measured to determine conformance to the physical interchangeability requirement of 3.16. When a listed dimension is not within the specified or design limits, it shall be considered a major defect.

a. External and internal dimensions of cases, covers, and insertable assemblies, when such dimensions affect mating parts.

b. Dimensions of cavities, when such dimensions affect insertion of items.

c. Location of fasteners on separable parts or assemblies which must mate, such as cases, covers and mountings.

d. Location of connectors, locking pins, fasteners, and mountings which receive mating parts of plug-in assemblies and major units, and location of mating parts on the plug-in assembly or major unit.

4.14 Quality conformance of preparation for delivery. - Preparation for delivery shall be inspected in accordance with Specification MIL-P-116 to determine conformance to the requirements of Section 5.

4.15 Rough handling test (preparation for delivery). - When rough handling test in accordance with Specification MIL-P-116 is required by the contract (See 6.2), the following functional test shall be conducted to determine freedom from operational malfunction caused by rough handling: 4.11 Operational Test.

5. Preparation for delivery. -

5.1 Preservation and packaging. - Preservation and packing shall be Level A or C as specified. (See 6.2)

5.1.1 Level A. -

5.1.1.1 Cleaning. - Each Coupler Antenna CU-942( )/AR, Coupler Antenna CU-1894( )/AR and Antenna AS-1703( )/AR shall be cleaned in accordance with process C-1 of MIL-P-116.

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5.1.1.2 Drying. - Each Coupler Antenna CU-942( )/AR, Coupler Antenna CU-1894( )/AR and Antenna AS-1703( )/AR shall be dried in accordance with the applicable procedure of MIL-P-116.

5.1.1.3 Preservation application. - None required

5.1.1.4 Unit packaging. - Unit packaging shall be in accordance with the methods prescribed in MIL-P-116 as specified herein.

5.1.1.4.1 Coupler Antenna CU-942( )/AR, Coupler Antenna CU-1894( )/AR . - Each coupler shall be individually packaged Method III as follows: Cushion each coupler on all surfaces with cells or pads or both conforming to PPP-F-320, W5c, weather resistant, designed and fabricated to protect all projections and absorb the shock of impact in handling and transit. Place each cushioned item within a close fitting fiberboard box conforming to PPP-B-636, W6c, weather reistant, box closure shall be as specified in the appendix of the box specification.

5.1.1.4.2 Antenna AS-1703( )/AR. - Each antenna shall be individually packaged Method III as follows: Cushion each antenna by wrapping in paperboard conforming to Type I, PPP-P-291, secure the wrap with gum kraft tape conforming to PPP-T-45.

5.1.2 Level C. - Coupler Antenna CU-942( )/AR, Coupler Antenna CU-1894( )/AR, and Antenna AS-1703( )/AR shall be packaged in a manner that will afford adaquate protection against corrosion, deterioration and damage during shipment from the supply source to the first receiving activity.

5.2 Packing. - Packing shall be Level A, B or C as specified. (See 6.2)

5.2.1 Level A. - A quantity of items packaged as specified in 5.1.1.4.1 and 5.1.1.4.2 bearing the same stock numbers shall be packed within a close-fitting fiberboard box conforming to PPP-B-636, Type CF, class weather resistant, special requirement use. Box closure shall be as specified in the appendix of the box specification. To facilitate palletization, fiberboard boxes shall be uniform in size and contain equal quantities of the packaged item to the greatest extent practicable.

5.2.1.2 Palletized load. - A quantity of containers packed as specified in 5.2.1 shall be placed on a pallet, load Type I, conforming to MIL-STD-147. A fiberboard cap shall be employed over the load having two sides extending down the stacked load at least 12 inches to accommodate marking requirements. The cap shall be fabricated of fiberboard conforming to PPP-F-320, class weather resistant, W5s. The load shall be "bonded" to the pallet by strapping.

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5.2.1.3 Less than palletized load. - When quantities per destination are less than a pallet load, containers specified in 5.2.1 shall be reinforced by pressure sensitive filament tape conforming to PPP-T-97, Type IV or specified in the appendix to the box specification no further packing shall be required.

5.2.2 Level B. - Coupler, Antenna CU-942( )/AR, Coupler Antenna CU-1894( )/AR, and Antenna AS-1703( )/AR, packaged as specified in 5.1, shall be packed as specified in 5.2.1.

5.2.3 Level C. - Coupler Antenna CU-942( )/AR, Coupler Antenna CU-1894( )/AR, and Antenna AS-1703( )/AR packaged as specified in 5.1, shall be packed as specified in 5.2.1 except that pallets, fiberboard boxes, and fiberboard caps shall be of the type size, commonly used for the purpose and shall comply with the rules and regulations of the command carrier as applicable to the mode of transportation. Packing shall be accomplished in a manner that will afford adequate protection from the supply source to the first receiving activity.

5.3 Marking. - In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. - Antenna AS-1703( )/AR and Coupler, Antenna CU-942( )/AR is used with radio sets for communications when installed in fixed and rotary wing aircraft. Antenna AS-1703( )/AR and Coupler, Antenna CU-1894( )/AR is used with radio sets for communications when installed in OH-6 aircraft.

6.2 Ordering data. - Procurement documents should specify the following:

a. Title, number and date of this specification and any amendment thereto.

b. Level of packaging and level of packing required for shipment (Level A, B or Level C).

c. The specific paragraphs of Section 5 which are applicable to the particular procurement.

d. Preproduction inspection:

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(1) Four (4) sample units of each item cited in Section 1 are generally required so that lengthy environmental tests can be completed on two sample units while complete performance measurements can be made on the other sample unit. (See 3.3)

(2) Preproduction pack(s) as follows:

(a) Makeup of pack(s)

(b) Number of each kind of pack to be submitted.

(c) Inspection to be performed thereon (including rough handling).

(d) Marking and shipping of samples.

(e) Place of final inspection.

(f) Quantity of spare parts required. (See 3.17)

(g) Submission of the statement of treatment referenced in 3.10 should be as specified in the solicitation/contract unless otherwise specified. This statement should be submitted to the Contracting Officer.

6.3 Nomenclature. - The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: CU-942A/AR. The contractor should apply for nomenclature in accordance with the applicable clause in the contract. (See 1.1)

6.4 Location of operational inspection. - It is desirable that the operational inspection (4.11) be performed at a location that will minimize handling (which might cause damage to the equipment) after this inspection is completed. It is recommended that the entire lot (including all previously inspected sample units) be inspected immediately prior to packaging.

6.5 Group C inspection. - Approval to ship may be withheld, at the discretion of the Government, pending the decision from the contracting officer on the adequacy or corrective action. (See 4.5.4.1)

6.6 Definition of level B. - Level B packaging requirements detailed herein are designed to furnish protection against corrosion, deterioration, and physical damage during domestic shipments; covered storage for a period of three years or less; and distribution within the continental limits of the United States.

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6.7 Dimensional data. - Dimensions of packaging materials in the Bill of Materials Table are based on the dimensions of the equipment cited in the table. When dimensions of the items very, the dimensions of the packaging materials shall be adjusted accordingly.

6.8 Verification inspection. - Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification by the Government will be adjusted to make maximum utilization of the contractors quality control system and the quality history of the product.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-C-55693(EL), COUPLERS, ANTENNA CU-942( )/AR, CU-1894( )/AR AND ANTENNA AS-1703( )/AR

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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